



IIT ROORKEE



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CERTIFICATION COURSE

Extension of H-O Theory

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Endowment Data

Figure 1.1 Capital stock per worker in 2014 in United States dollars (PPP 2011)

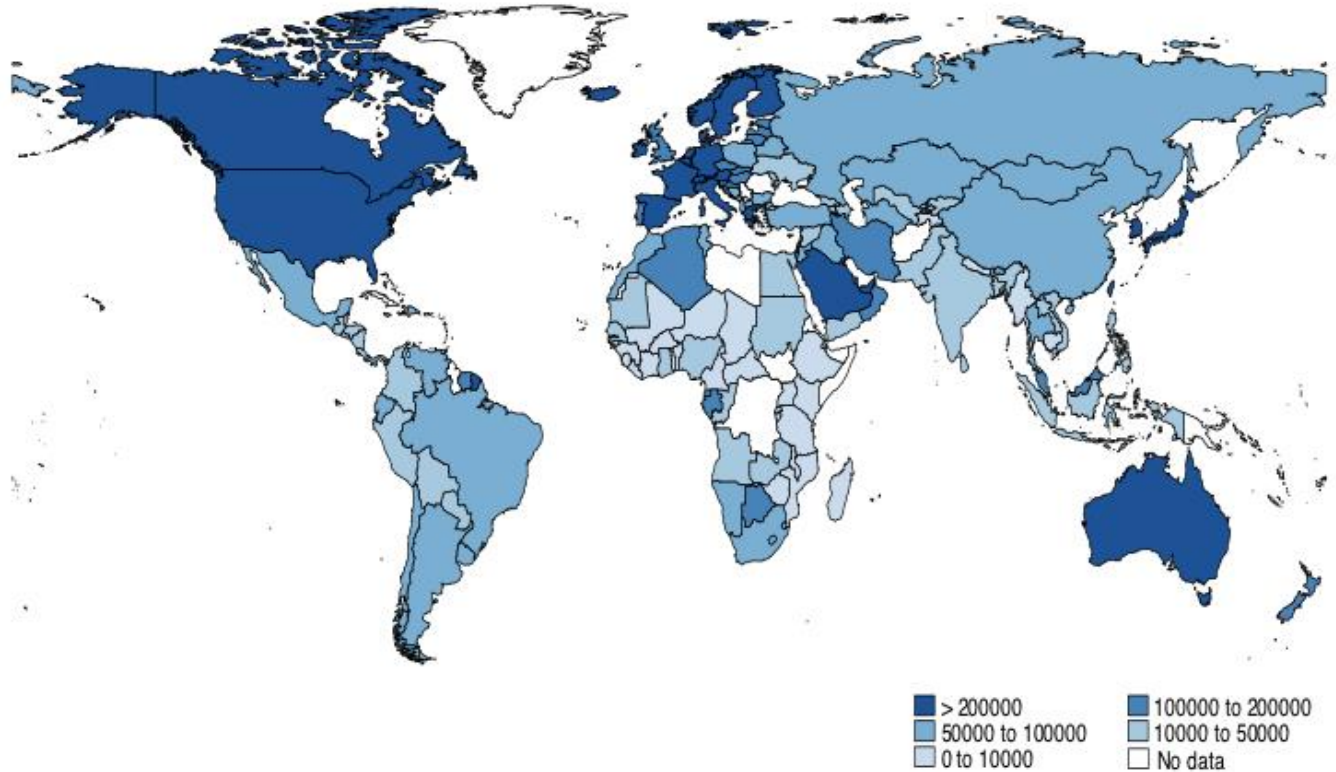


Figure 1.2 Percentage change of physical capital (1994- 2014)

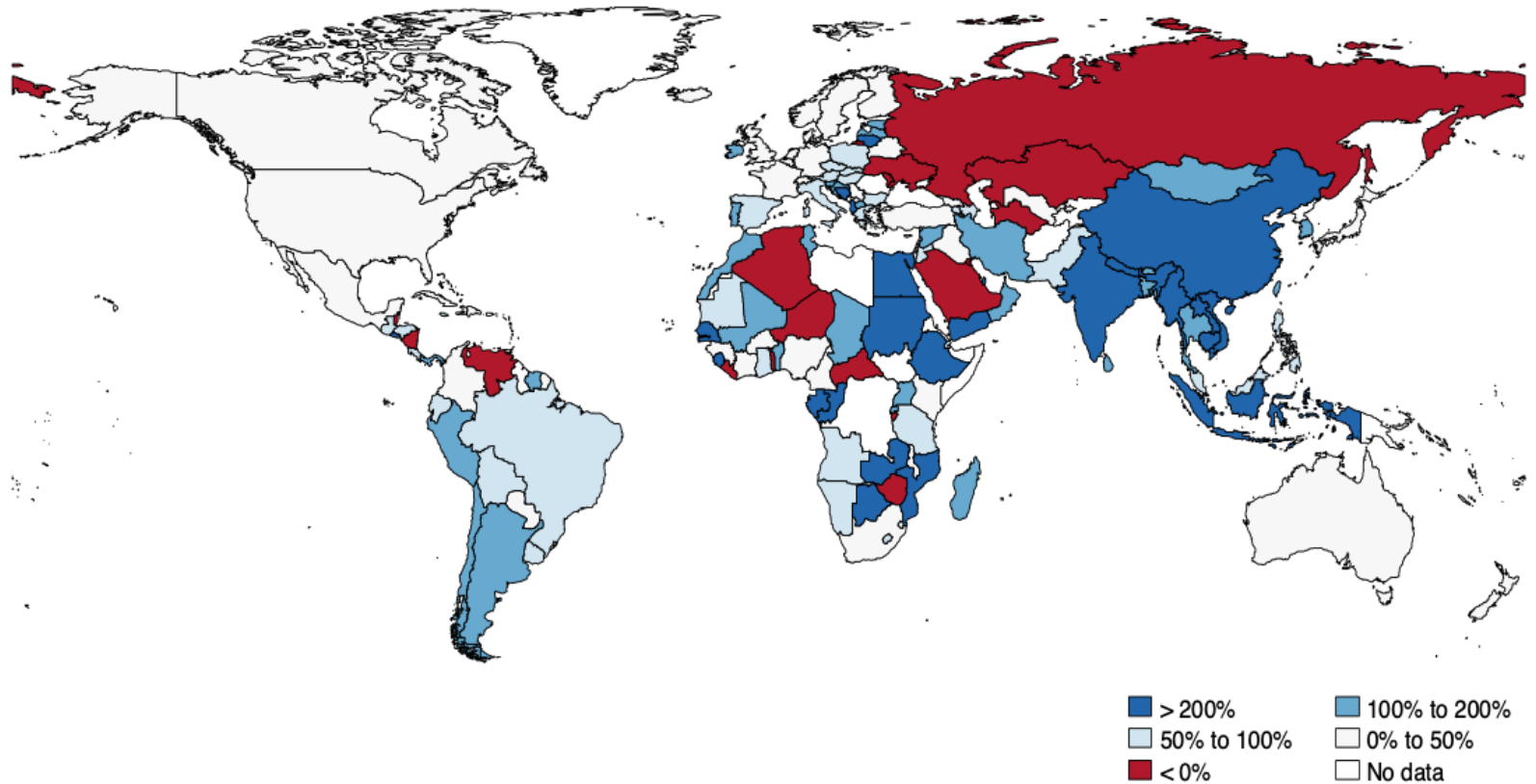
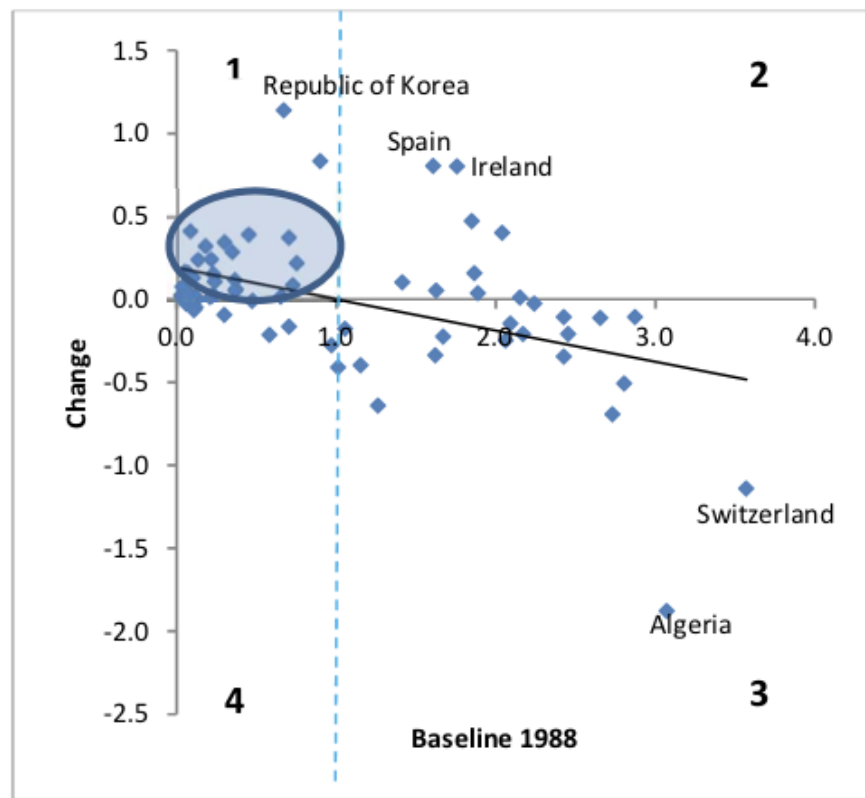


Figure 2. Relative changes in physical capital stock (1988-2014)

(a): Physical Capital (all countries)



(b) Physical capital (emerging economies)

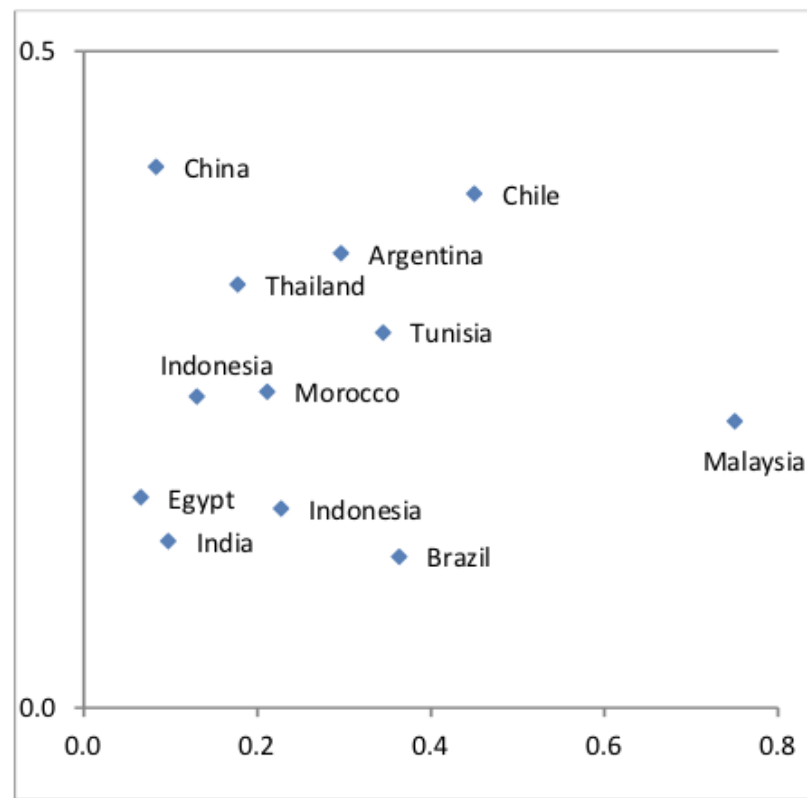
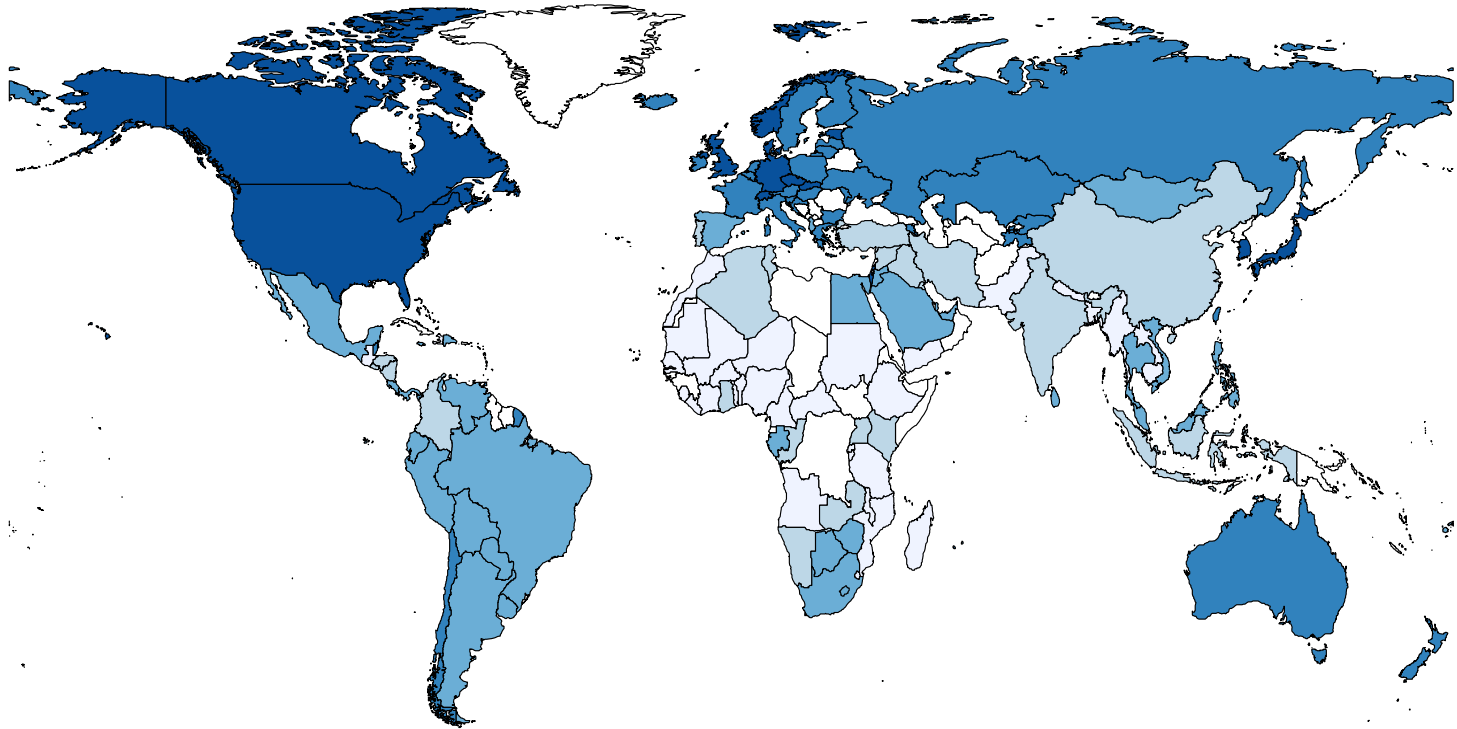


Figure 3.1 Human capital index per person in 2014



Human capital is measured as a composite of average schooling years and the return on education.

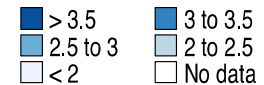
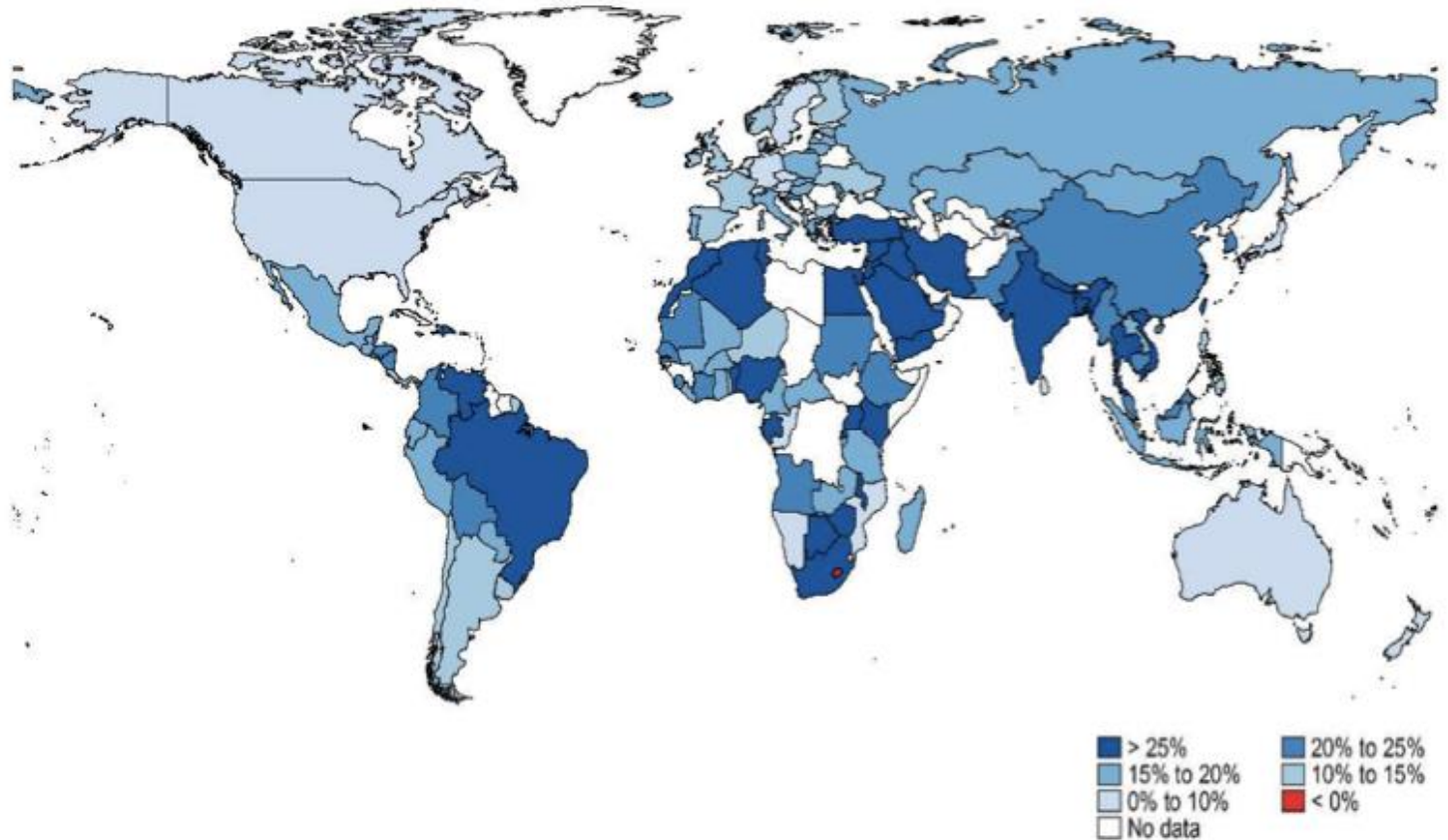


Figure 3.2 Percentage change of human capital (1994-2014)



Extensions of the HO Model

- There are several alternative trade models that elaborate on the theory of comparative advantage
 - **Product cycle model** – focuses on the speed of technological change and life history of many manufactured items through periods of innovation, stabilization, and standardization
 - **Intra-firm trade model** – allows for comparative advantage but incorporates industrial organization

Trade Based on Dynamic Technological Differences

- **Technological Gap Model (*Posner, 1961*)**
 - Great deal of trade due to new products and production process
- **Product Cycle Model (*Vernon, 1966*)**
 - Advanced industrialized countries develop and introduce new products, with temporary monopoly power as the sole exporter of the product.
 - As the technology producing the product becomes more widespread, production will spread to other nations.
 - This moves international trade to a standard comparative advantage framework



Trade Based on Dynamic Technological Differences

- **Product Cycle Model** (*Vernon, 1966*)
 - As production becomes standardized, the original introducer of the product loses its technologically based comparative advantage in the production of the product and becomes an importer of the product.



Product Cycle

- Developed by Raymond Vernon
- Argument: Production of a good is cyclical
 - When a manufactured good is developed, producers experiment and seek consumers' reactions
 - When production leaves the early stage, the good begins to be standardized in terms of size, features, and manufacturing process
 - Finally, consumption of the good in a high-income country exceeds its production: production moves where labor costs are lower

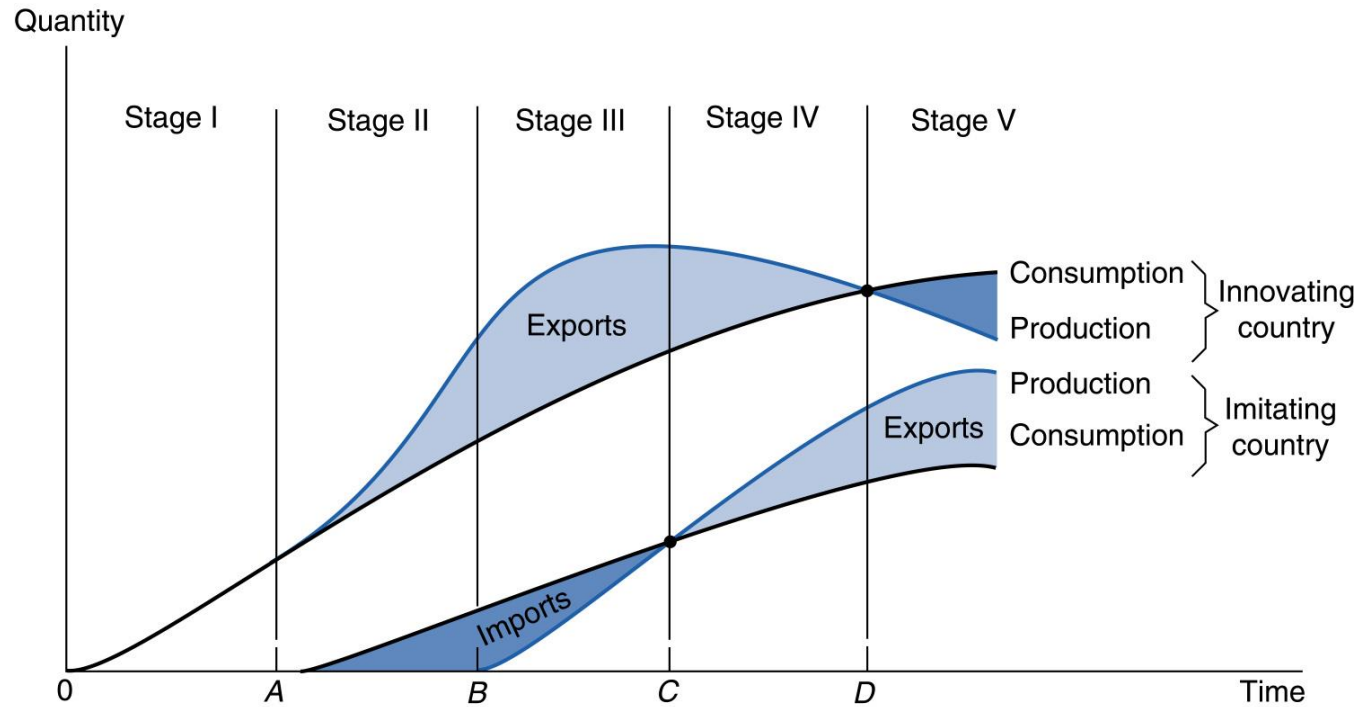
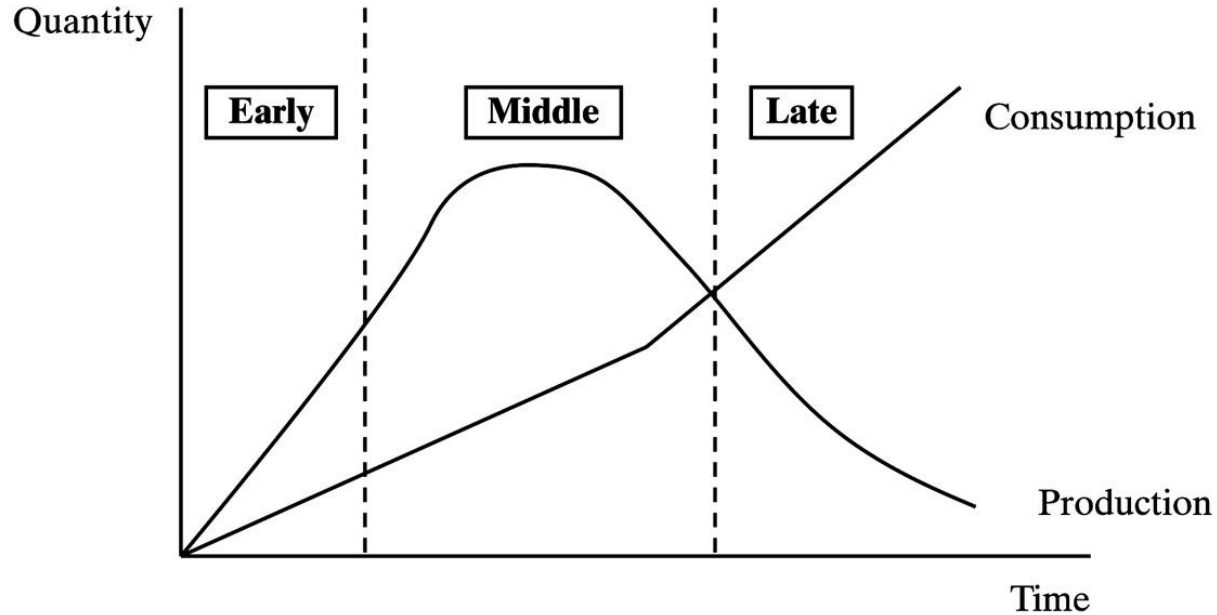
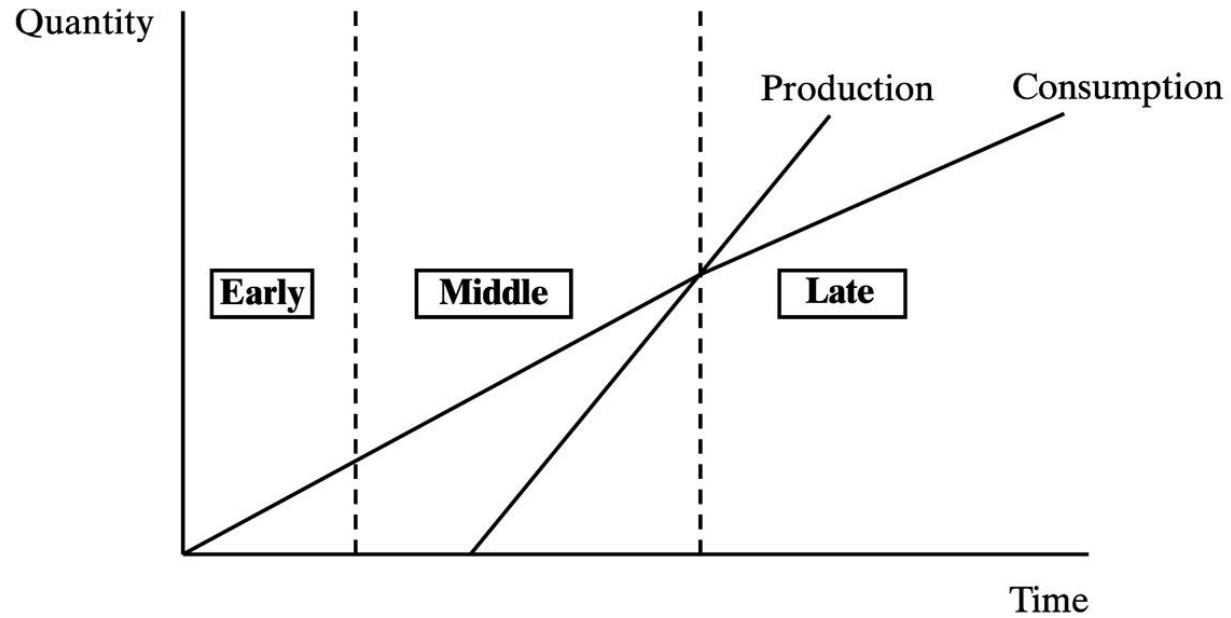


FIGURE 6-4 The Product Cycle Model.

Product Cycle (cont.)





Outsourcing and Intra-Firm Trade

- Much of international trade is **intra-firm trade** – trade between the parent company in the home country and its affiliate in a foreign country
- Reasons for intra-firm trade
 - Firms take advantage of cross-country differences in the prices of inputs
 - A firm may obtain cheaper and better inputs through its foreign affiliate rather than independent foreign firms
 - Similarly, a firm may reduce distribution costs in a foreign market by operating through an affiliate



Intra-Firm Trade (cont.)

- Intra-firm trade is growing in importance
 - In the mid-1990s, 2/3 of U.S. merchandise exports and 2/5 of U.S. merchandise imports carried out within firms
- Intra-firm trade may have important economic benefits
 - Expansion of multinational corporations (MNCs) helps diffuse technology across national borders



Outsourcing v. Intra-Firm Trade

- An alternative to intra-firm trade is outsourcing: arms-length transactions to provide inputs and/or processing.
- Many firms do both outsourcing and intra-firm trade.
- More broadly, we can refer to *globalization of production*.



Implications of Outsourcing and Intra-Firm Trade

- Outsourcing looks like technological change in the data.
 - Thus, empirical results on the Stolper-Samuelson theorem may need to be rethought.
 - In addition, clean theoretical results in an environment with outsourcing are hard to come by (*dimensionality problems*).
- Outsourcing in models with distortions (unions or welfare states) may produce large welfare effects.



Costs of Transportation, Environmental Standards and International Trade

- **Transportation costs**

- **Transport, or logistics, costs** are the freight charges, warehousing costs, costs of loading and unloading, insurance premiums, and interest charges incurred while goods are in transit between nations.
- Homogeneous goods will be trade internationally only if the pretrade price difference exceeds transport costs.



Costs of Transportation, Environmental Standards and International Trade

- **Transportation costs**

- **Nontraded goods and services** are goods for which transport costs exceed price differences across nations.
- Examples:
 - Cement is not traded internationally because of its high weight-to-value ratio.
 - Average people do not travel from New York to London for a haircut.



Costs of Transportation, Environmental Standards and International Trade

- **Two ways to analyze transport costs**
 - General equilibrium analysis
 - Uses production frontiers or offer curves, and expresses transport costs in terms of relative commodity prices.
 - Partial equilibrium analysis
 - Analyze absolute cost by holding constant exchange rates, income, and all else in the two nations except amount of good produced, consumed and traded.
 - More straightforward method than general equilibrium analysis.



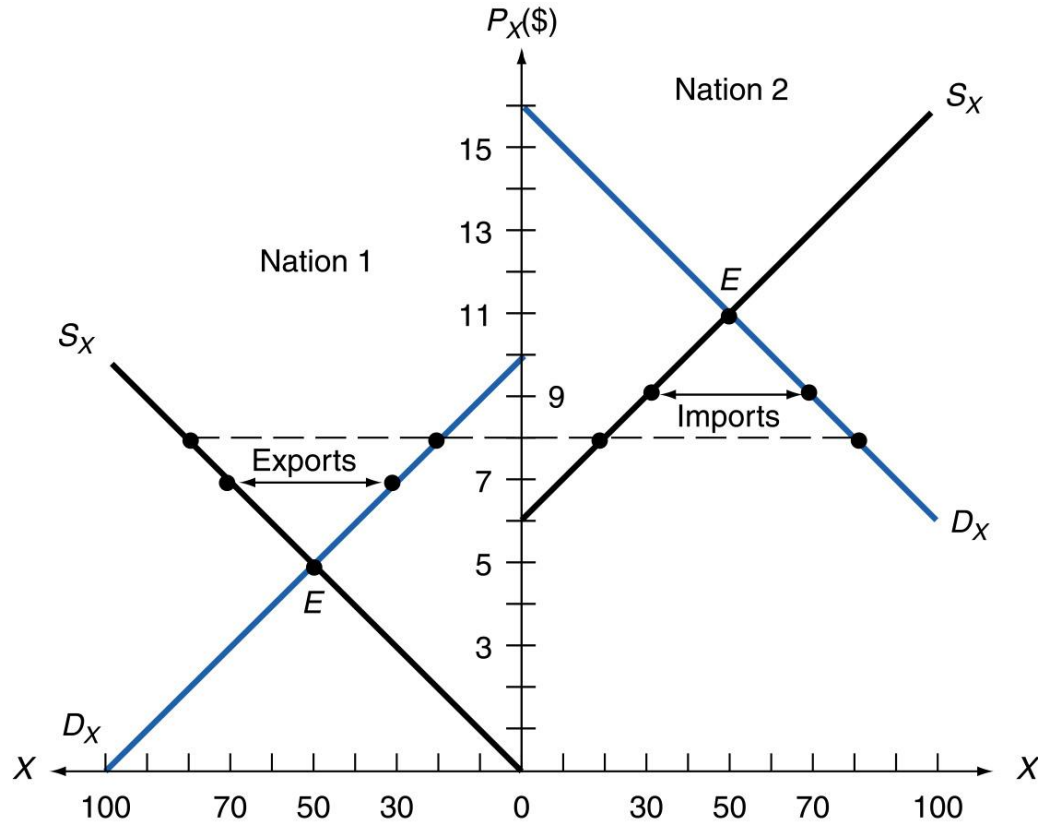
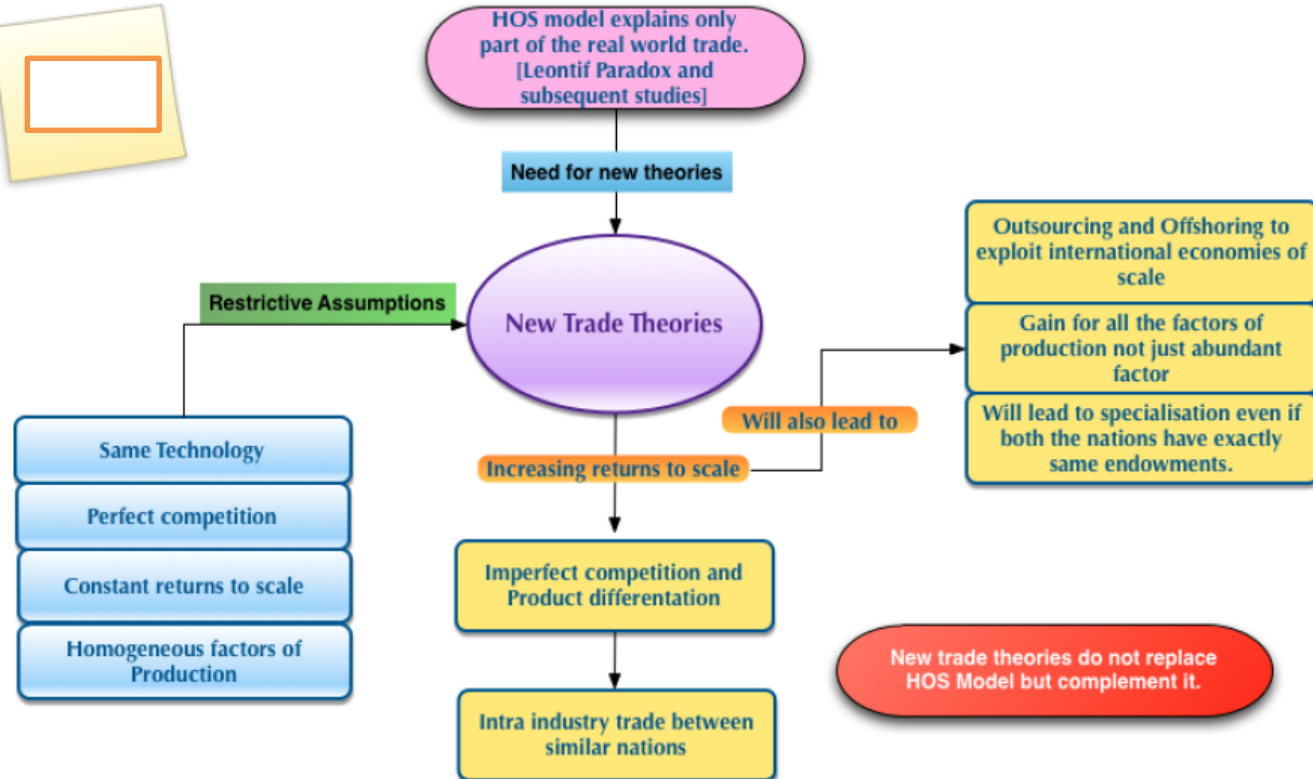


FIGURE 6-5 Partial Equilibrium Analysis of Transport Costs.



*Thank
you*

